

AWS Certified AI Practitioner (AIF-C01)

Ultimate AI/ML Services Ultimate Cheat Sheet & Exam Pass Guide

1. THE PILLARS OF AWS GENAI & MACHINE LEARNING

The exam strictly tests your ability to distinguish between serverless Foundation Model architectures and managed custom lifecycle training platforms.

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Bedrock	A fully managed, serverless service that exposes high-performing third-party & AWS Foundation Models (FMs) (Claude, Llama, Mistral, Command, Nova) via a single unified API endpoint. No infrastructure to manage.	• Serverless application deployment • No ML experience required • Customizing existing models using Fine-Tuning or Retrieval-Augmented Generation (RAG) • Continuous pre-training on unlabeled data
Amazon SageMaker AI	A comprehensive, modular platform for data scientists to build, train, optimize, and deploy custom ML models from scratch using arbitrary open-source libraries or custom algorithms.	• Total infrastructure Granular Control • Creating custom training jobs or processing pipelines • Hyperparameter optimization (HPO) tuning • Deep feature engineering, artifact logging, and container instances

High-Yield Bedrock & SageMaker Sub-Features (Must Know)

Amazon Bedrock Guardrails: Implements governance and user safeguards. Triggers: Responsible AI , filtering toxic input, stripping Personally Identifiable Information (PII) , preventing hallucinations, and creating model safety alignment boundaries.
SageMaker Canvas: A visual graphical UI tool. Triggers: Build ML models and run predictions without writing a single line of code ; visual workflow for business analysts.
SageMaker Clarify: Bias and explainability service. Triggers: Checking datasets for pre-training/post-training bias , answering why a model reached a specific decision (explainability), drift alerts.
SageMaker JumpStart: An internal library hub. Triggers: Quickly deploy popular pre-trained models or architectures directly inside private SageMaker environment boundaries.

2. AWS GENERATIVE AI ASSISTANTS & AGENTS

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Q Business	An AI conversational assistant that links to internal organizational data stores to index files, synthesize answers, and construct textual materials securely.	• Enterprise knowledge assistant • Indexing corporate wikis, internal S3, Slack, or Salesforce • Answering employee questions regarding internal business policy documents

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Q Developer	An AI assistant tuned directly for technical engineering pipelines. Integrates with your command line and your IDE interface to modify, refactor, generate, or query programming files.	- AI coding companion - Generating automatic unit test coverage blocks - Scanning software lines for security flaws - Explaining legacy architectural frameworks (e.g., old Java structures)
Amazon CodeWhisperer	The legacy name for the IDE-based code generation tooling.	<i>Note: If listed, it is now part of Amazon Q Developer. Focus on Q Developer for coding contexts.</i>

3. PRE-TRAINED LANGUAGE, TEXT, & EXTRACTION SERVICES

These are managed turnkey services. They require ****zero ML training**** or model deployment; you simply pass data to an API.

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Comprehend	Advanced Natural Language Processing (NLP) designed to unearth semantic intelligence from raw, messy text corpora.	- Tracking sentiment analysis (positive/negative/neutral feedback) - Extracting key physical entities, syntax groups, or locations - Redacting sensitive personal variables automatically from logs
Amazon Textract	An intelligent extraction program that converts scanned documents or image layouts into digital machine-readable text arrays.	- Pulling structured keys, values, tables, and form inputs from PDFs - Isolating lines from medical claims, invoices, or manual logbooks - Processing handwritten data fields without human entries
Amazon Translate	A fast neural machine translation system optimized for multi-language localizations.	- Converting standard asset blocks into localized regional scripts Real-time chat stream translation for international service desks
Amazon Kendra	An intelligent corporate search platform powered behind the scenes by ML to scan siloed indexing systems using natural queries.	Natural language semantic search index - Finding the target document matching user queries across isolated file shares <i>Exam Trap: Kendra locates documents; Q Business explicitly chats/summarizes them.</i>

4. AUDIO, VIDEO, & CONVERSATIONAL AI SERVICES

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Transcribe	Converts speech streams and audio payloads directly into formatted text documents via Automatic Speech Recognition (ASR).	Speech-to-text conversion - Generating structural timestamps or closed captions for videos - Processing phone recording payloads from call center storage buckets

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Polly	Converts written strings and articles into hyper-realistic, streaming vocal files via Text-to-Speech (TTS) architecture.	• Text-to-speech engine • Synthesizing spoken audio output for informational applications • Utilizing Speech Synthesis Markup Language (SSML) for pacing/breathing
Amazon Lex	A tool to engineer transactional interfaces using voice or chat inputs. Powered by the deep engine that drives Amazon Alexa.	• Building automated conversational chatbots • Managing interactions using slots, intents, and prompts • Front-ending Level-0 support call routing paths automatically
Amazon Rekognition	A computer vision pipeline that automatically isolates items, face models, activities, tracking tags, text lines, or objects across images/videos.	• Content moderation (flagging explicit/unsafe user uploads) • Biometric facial detection, confirmation, or indexing • Identifying specific items, text plates, or physical hazards in clips

5. PREDICTIVE ANALYTICS & SPECIALIZED DECISION TOOLS

Service Name	What It Does (Core Architecture)	Exam-Triggers & High-Yield Keywords
Amazon Personalize	A recommendation system powered by deep learning frameworks that tracks user paths and interactions to generate individualized product arrays.	• Real-time digital personalization engine • Building Amazon-style "Frequently bought together" or recommendation cards • Altering retail app displays based on historical click patterns
Amazon Forecast	A statistical prediction engine engineered to process complex historical time-series datasets to map future numerical expectations.	• Time-series data forecasting • Estimating upcoming inventory demand metrics, power draw, or parts usage • Anticipating seasonal workforce resource demands accurately
Amazon Fraud Detector	A highly automated tool that evaluates transactions or account events to isolate malicious actions or fraudulent entries.	• Catching rogue charge attempts, fake review setups, or bot account creations • Risk scoring online checkouts immediately based on historical markers

Critical Exam-Day Architecture Decision Rules:

- **Rule 1:** If the problem description asks for an architecture requiring "**no ML expertise**," "**pre-trained models**," or "**minimal operational overhead**," choose the specific specialized AI service (Comprehend, Rekognition, Textract, Translate, Bedrock).
- **Rule 2:** If the problem calls for "**complete infrastructure customization**," "**training custom open-source scripts**," "**tuning hyperparameters**," or logging custom metadata variables, your destination is always **Amazon SageMaker AI**.
- **Rule 3:** If the prompt involves scanning enterprise internal files to build a generative conversational interface, look for **Amazon Q Business**. If it is only keyword index lookup, select **Amazon Kendra**.